

Wuao Liu

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Education

University of Massachusetts Amherst

Ph.D. in Computer Science

Amherst, MA

Aug 2024 – Dec 2028 (expected)

University of Michigan, Ann Arbor

M.S. in Robotics

Ann Arbor, MI

Aug 2021 – Apr 2023

Zhejiang University (ZJU)

B.Eng. in Robotics

Hangzhou, China

Sep 2017 – Jun 2021

Work Experience

Honda Research Institute

Research Scientist Intern (Multimodal ML)

San Jose, CA

May 2025 – Aug 2025

- Built a multimodal framework integrating spatial audio cues with large language models for AR/VR scene understanding.
- Designed a spatial audio encoder jointly optimized for sound classification and direction-of-arrival (DoA) estimation.
- Implemented spatial audio data generation and filtering pipelines for a large-scale dataset with **872k** perception and reasoning question-answer pairs.

University of Michigan, Ann Arbor

Research Engineer

Ann Arbor, MI

Jun 2023 – Apr 2024

- Designed an egocentric video collection protocol and curated a dataset of **224** instructional cooking videos.
- Trained a Faster R-CNN object detector for **14** kitchen items, achieving **85.99** mAP on the curated video dataset.
- Developed a procedural step classification model achieving **92.54%** accuracy, enabling automated detection of step-transition errors.

Research Experience

University of Massachusetts Amherst

Self-supervised Learning for Natural Sounds

Advisor: Prof. Grant Van Horn

Jan 2025 – May 2025

- Extended the Audio Masked Autoencoder (AudioMAE) pretraining framework to wildlife sounds.
- Enabled self-supervised representation learning via audio spectrogram reconstruction and improved classification accuracy on the iNatSounds dataset by **1.6%**.

University of Massachusetts Amherst

Contrastive Learning for Audio Geolocation

Advisor: Prof. Grant Van Horn

Aug 2024 – Dec 2024

- Proposed a novel framework for global-scale geolocation using natural sounds.
- Developed a retrieval-based CLIP model, integrating audio geolocation with species range prediction.
- Achieved a median localization error of **1127** km and regional-level accuracy of **17.2%** on the iNatSounds dataset.

Publications

M. Chasmai, **Wuao Liu**, S. Maji, G. Van Horn. *Audio Geolocation: A Natural Sounds Benchmark*. Under review. [\[paper\]](#)

J. Zhong, Y. Lai, M. Li, Y. Li, **Wuao Liu**, et al. *From Shortcuts to Reasoning: Robust Post-Training of Theory of Mind with Reinforcement Learning*. Under review.

F. Bellos, Y. Li, **Wuao Liu**, J. J. Corso. *Can Large Language Models Reason About Goal-Oriented Tasks?* In *ACL Workshop on the Scaling Behavior of Large Language Models*, 2024. [\[paper\]](#)

Technical Skills

Programming: Python; C++; MATLAB; $\text{\LaTeX}$; ROS

ML/Multimodal: PyTorch; TensorFlow2; HuggingFace Ecosystem; verl